INSTITUTO COLLEGE

MATH 110: General Education Math

Fall 2026

Homework 3

Name: _____

1. (*6 pts*)

   (a) Evaluate $\sum_{i=1}^{4}(3i+1)$.

   (b) Write $5 + 10 + 15 + \ldots + 50$ in sigma notation.

2. (*6 pts*) Two fair dice are rolled. Let E be the event "the product of the numbers the dice land on is 12." Write E by listing its sample points, and find $P(E)$.

3. (*8 pts*) A patient portal assigns each patient a random 4-digit PIN, using the digits 0 through 9. Digits may be repeated, and every PIN is equally likely. What is the probability that a randomly assigned PIN contains at least one 0?

4. (*8 pts*) A rapid strep test comes back positive for 20% of the patients tested at a clinic. Two unrelated patients are tested, and their results are independent.

(a) What is the probability that both patients test positive?

(b) What is the probability that exactly one of the two patients tests positive?

5. (*10 pts*) A nurse will choose 2 of her 5 patients, Ana, Ben, Carla, David, and Eva, to take part in a follow-up survey. Every pair of patients is equally likely to be chosen.

(a) (*2 pts*) Write the sample space by listing its sample points. Use first initials, so that AB means Ana and Ben are chosen.

(b) (*1 pt*) Check your answer to part (a) using a combination number.

(c) (*2 pts*) Let A be the event "Ana is chosen." List the sample points of A, and find $P(A)$.

(d) (*1 pt*) Let B be the event "neither Ana nor Ben is chosen." List the sample points of B, and find $P(B)$.

(e) (*2 pts*) Explain why A and B are mutually exclusive, and use the Addition Rule to find $P(A \cup B)$.

(f) (*2 pts*) Describe the event $(A \cup B)^c$ in words, and use your answer to part (e) to find its probability.

6. (*10 pts*) A patient is chosen at random from a clinic's records, and the number of emergency room visits the patient made last year is recorded. The sample space is $S = \{0, 1, 2, 3, 4\}$, where 4 means 4 or more visits. It is known that

$$P(0) = 0.55, \quad P(1) = 0.25, \quad P(2) = 0.12, \quad P(3) = 0.05.$$

(a) (*2 pts*) Write out $\sum_{x \in S} P(x)$ term by term. What must this sum equal? Use your answer to find $P(4)$.

(b) (*2 pts*) Let $A = \{2, 3, 4\}$ be the event "at least 2 visits." Find $P(A)$.

(c) (*2 pts*) Describe the event A^c in words, and find $P(A^c)$ in two ways: using complements, and by adding the probabilities of its sample points.

(d) (*2 pts*) Are the events $\{0\}$ and A mutually exclusive? Explain.

(e) (*2 pts*) A student says: "There are 5 sample points, so each one has probability $\frac{1}{5}$." What is wrong with this reasoning?

7. (*12 pts*) At a hospital, 35% of the patients are over age 65, 25% have diabetes, and 12% are over age 65 and have diabetes. A patient is chosen at random. Let O be the event "the patient is over age 65" and D the event "the patient has diabetes."

(a) (*2 pts*) Use the Inclusion-Exclusion Principle for Events to find $P(O \cup D)$.

(b) (*2 pts*) Find $P\big((O \cup D)^c\big)$. Describe this event in words.

(c) (*3 pts*) Find the probability that the patient is over age 65 but does not have diabetes.

(d) (*2 pts*) Are O and D mutually exclusive? Explain.

(e) (*3 pts*) Are O and D independent? Explain.

8. (*14 pts*) Two fair dice are rolled. Let A be the event "the first die lands on an even number," B the event "the numbers the dice land on add up to 7," and C the event "at least one die lands on 5."

(a) (*2 pts*) Use the Fundamental Principle of Counting to find $|A|$, and find $P(A)$.

(b) (*3 pts*) Find $|B|$ and $|C|$, and find $P(B)$ and $P(C)$. (You do not need to list the sample points of C.)

(c) (*3 pts*) List the sample points of $A \cap B$. Are A and B independent? Explain.

(d) (*3 pts*) Find $|A \cap C|$, and use the Inclusion-Exclusion Principle for Events to find $P(A \cup C)$.

(e) (*3 pts*) Are A and C independent? Explain.

9. (*14 pts*) A bag contains 3 red marbles and 2 blue marbles, labeled R_1, R_2, R_3, B_1, and B_2. Two marbles are drawn at random, one after the other, and the order is recorded. Let X be the event "the first marble is red" and Y the event "the second marble is red."

(a) (*2 pts*) Suppose the first marble is put back in the bag before the second is drawn. How many sample points does the sample space have?

(b) (*3 pts*) Still putting the first marble back, find $|X|$, $|Y|$, and $|X \cap Y|$, and find $P(X)$, $P(Y)$, and $P(X \cap Y)$.

(c) (*2 pts*) Are X and Y independent when the first marble is put back? Explain.

(d) (*4 pts*) Now suppose the first marble is not put back. How many sample points does the sample space have? Find $|X|$, $|Y|$, and $|X \cap Y|$, and find $P(X)$, $P(Y)$, and $P(X \cap Y)$.

(e) (*3 pts*) Are X and Y independent when the first marble is not put back? Explain why your answer makes sense.

10. (*12 pts*) A fair coin is tossed 3 times. Let A be the event "at least two heads," B the event "the first toss lands on heads," and C the event "exactly one head."

(a) (*3 pts*) List the sample points of A, B, and $A \cap B$, and find $P(A)$, $P(B)$, and $P(A \cap B)$.

(b) (*3 pts*) Use the definition of conditional probability to find $P(A \mid B)$ and $P(B \mid A)$. Compare $P(A \mid B)$ with $P(A)$, and $P(B \mid A)$ with $P(B)$.

(c) (*3 pts*) List the sample points of C and $B \cap C$. Use the definition of conditional probability to find $P(C \mid B)$, and compare it with $P(C)$.

(d) (*3 pts*) Are A and B independent? Are B and C independent? Explain.